INCH-POUND

MIL-N-85615/6
15 December 1989

MILITARY SPECIFICATION SHEET

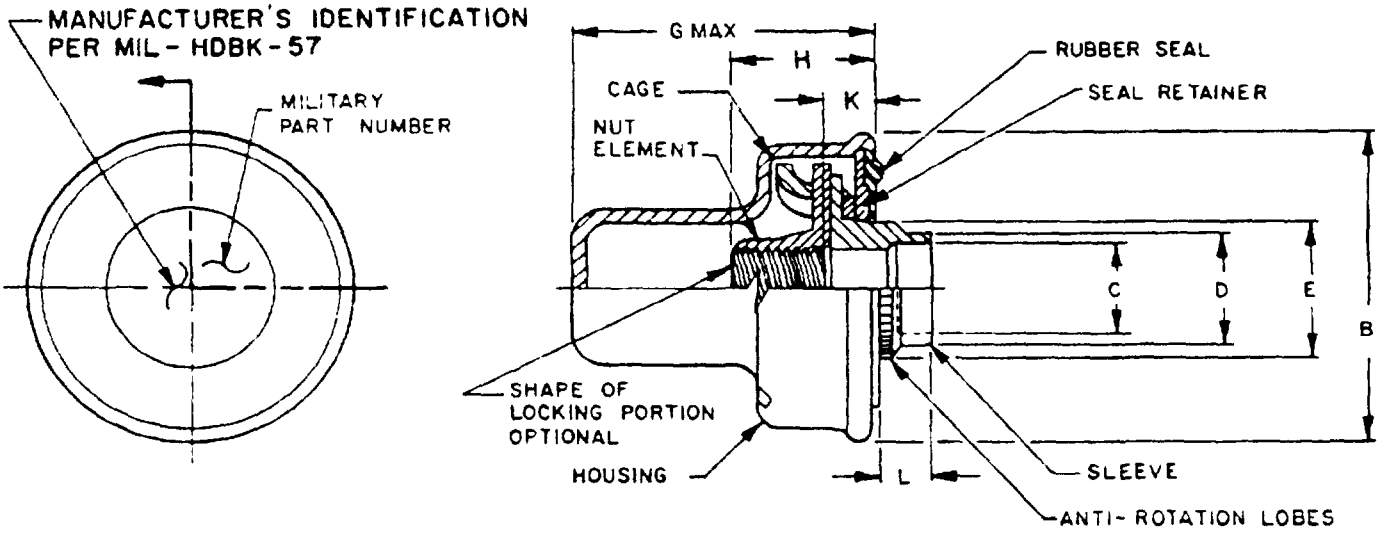
NUT ASSEMBLY, SELF-SEALING, SHEAR SLEEVE, CLOSE TOLERANCE, SELF-LOCKING, REDUCED DIAMETER SLEEVE, 125 KSI FTU, 225°F, VARIABLE NUT HEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-N-85615.

REQUIREMENTS:

Qualification required.



K = Thread relief, sleeve flange, cage and seal thickness
H, K = Nut height, see Table IV; L = Grip length, see Table III

FIGURE 1. Standard height self-sealing nut assembly (close tolerance).

AMSC N/A FSC 5310
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TABLE I.

Code	Thread	B Dia Max	C Dia Ref	D Dia Max	E Ref	G Max	J Ref	Axial Lbs Min
08	.1640-32UNJC-3B	.565	.185	.205	.219	.562	.167	1720
3	.1900-32UNJF-3B	.655	.211	.231	.245	.625	.192	2460
4	.2500-28UNJF-3B	.780	.271	.291	.305	.760	.252	4580

1. Application criteria. The nuts shall be used in applications where a close tolerance, self-sealing nut is required to eliminate leak paths in fuel tank access covers or other applications requiring environmental sealing; in pressure compensated areas or to seal against vapors, gases and liquids compatible with the Buna-N rubber seal material. MIL-N-85615/6 nuts are intended to be used in applications subject to shear loading where joint rigidity is a requirement.

2. Oversize nuts. MIL-N-85615/7 (1/64" oversize lobe nut assemblies) may replace MIL-N-85615/6 nuts for repair operations.

3. Military installation and removal. Military installation and removal shall be in accordance with MIL-HDBK-271-103.

4. Materials.

   4.1 Nut element. A: Steel AISI 1050 per AMS 5085 or AISI 4130 per AMS 6351.
   C: CRES A-286 per AMS 5525.

   4.2 Sleeve and housing. CRES AISI 304 per AMS 5639 or AISI 305 per AMS 5514.

   4.3 Cage. A: Steel AISI 1050 per AMS 5085.
   C: CRES 17-7PH per AMS 5528.

   4.4 Seal retainer. Steel AISI 1010 per AMS 5047.

   4.5 Seal. Nitrile rubber (Buna-N) per MIL-R-6855, Class 1, Grade 60.

5. Heat treatment.

   5.1 Nut element. Rockwell C43 max.

   5.2 Cage. Rockwell C 40-47.

   5.3 Sleeve, housing and seal retainer. None.

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6. Coating.

6.1 Sleeve, cage and seal retainer. Cadmium plate per QQ-P-416, type II, class 2.

6.2 Nut element. Cadmium plate per QQ-P-416, type and class optional provided the element meets the salt spray requirements of MIL-N-25027.

6.3 Housing. Passivate per QQ-P-35.

7. Lubricant. Nut element only shall be dry film lubricated per MIL-L-46010, Type I.

8. Surface texture. Unless otherwise specified, the surface roughness shall not exceed 125 microinches measured in accordance with ANSI B46.1.

9. Nut float. Minimum radial float of .020 inch. The nut element shall be capable of engagement with bolt in maximum misaligned position.

10. Screw shank clearance within the sleeve = .006 max.

11. Workmanship.

11.1 Break all sharp edges and remove all burrs and slivers.

11.2 Installation. When installed into properly prepared holes, using proper tooling, the sleeve shall be capable of flaring smoothly with no visual evidence of cracking or splitting.

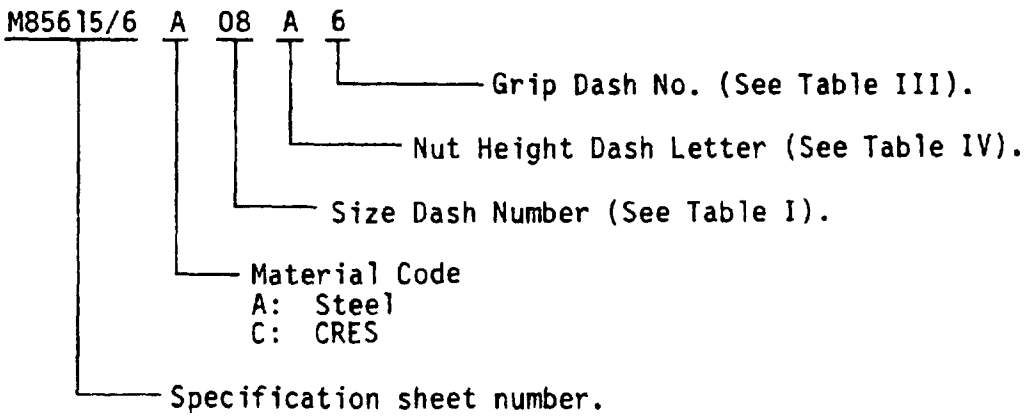
11.3 Threads shall be per MIL-S-8879 after plating and before lubrication.

12. Dimensions and tolerances.

12.1 All dimensions are in inches.

12.2 Tolerances. Decimals ±.010, angles ±1/2° unless otherwise specified.

13. Example of military Part or Identifying Number (PIN):



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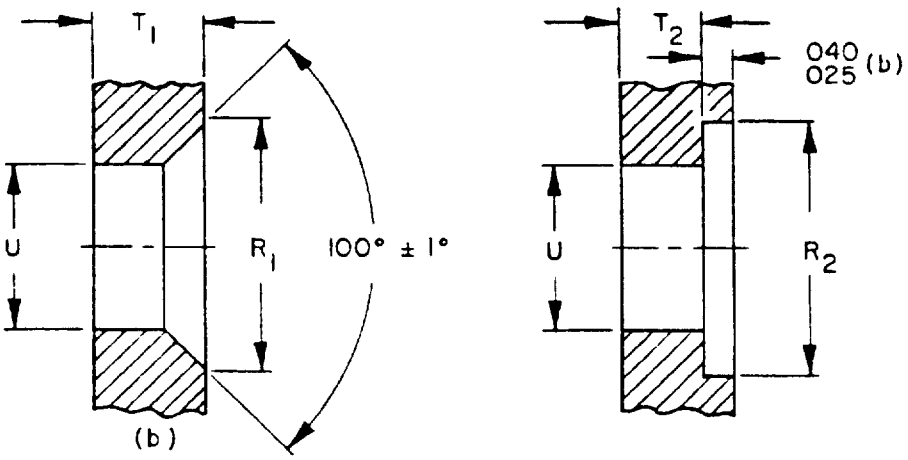
14. Administrative and contractual provisions.

14.1 For design purposes and in case of conflict this document takes precedence over documents referenced herein.

14.2 Referenced documents shall be of the issue in effect on the date of invitations for bids or request for proposal, except that referenced adopted industry documents shall give the date of the issue adopted.

14.3 The A.R.E. nut assemblies identified herein are proprietary products of Deutsch Fastener Corporation and are covered by the following patents:

U.S. 3,695,324 (expiring October 3, 1989); 3,765,078 (expiring October 6, 1990); 4,227,561 (expiring October 10, 1997); France 72,11463 (expiring April 13, 1990); Germany 22 17 490 (expiring March 31, 1992). The government does not have a royalty free license.



(b) COUNTERBORE AND COUNTERSINK ARE OPTIONAL TO BE USED IN APPLICATIONS REQUIRING SURFACE FLUSHNESS

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TABLE II. Hole preparation.

Size Dash Number	(a) Installation Hole Limit		R_1 ±.005	R_1 +.005 (-2 Grip) Only	R_2 +.020 -.005
	Max	Min			
08	.2115	.2085	.302	.258	.322
3 (e)	.2375	.2345	.328	.300	.348
4 (c)	.2975	.2945	.388	N/A	.408

TABLE III. Structure grip thickness table.

Grip Dash Number	T_1 Grip Thickness (Countersink)	T_2 Grip Thickness (Counterbore)	L Length Max
2 (c)	.055-.073 (c)	N/A	.073
4	.074-.106	.031-.062 (c)	.099
6	.107-.133	.063-.094	.130
8	.134-.165	.095-.125	.161
10	.166-.196	.126-.156	.192
12	.197-.227	.157-.187	.223
14	.228-.258	.188-.218	.254
16	.259-.289	.219-.250	.285
18	.290-.320	.251-.281	.316
20	.321-.351	.282-.312	.347
22	.352-.382	.313-.343	.378
24	.383-.413	.344-.374	.409
26	.414-.444	.375-.405	.440
28	.445-.475	.406-.436	.471
30	.476-.506	.437-.467	.502

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TABLE IV. Nut height.

Height Dash Letter	H Ref Size Dash Number		K Ref Size Dash Number	
	08, 3	4	08, 3	4
A	.308	.362	.118	.139
B	.370	.425	.176	.237
C	.432	.487	.239	.300
D	.494	.549	.302	.363

TABLE V. Nut weight.

Height Dash Letter	Weight lb/100 (d) Nut Size Dash Number		
	08	3	4
A	1.07	1.08	1.58
B	1.11	1.12	1.64
C	1.15	1.16	1.70
D	1.19	1.20	1.76

(a) Installation: Maximum and minimum hole limits can be reduced by .009 inch for size dash no. 08, 3 and 4 when installing in soft materials, including aluminum alloys with hardness not exceeding BHN 110 or Rockwell B62. Special bolt preparation may be required when installing this nut into structural materials having a hardness greater than Rockwell C35.

(b) Counterbore and countersink are optional; to be used in applications requiring surface flushness.

(c) Do not install size dash no. 4 on grip thickness less than .060 in.

(d) To obtain weights for different grip dash no., add increments of .020 lbs/100 for sizes 08, 3 and 4.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Review activity:
Army - AV, MI
Navy - OS, SH
DLA - IS

Preparing activity:
Navy - AS
(Project 5310-1650)

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